Quirks in ColliderML: simulation at scale and first ML tracking

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quirkml status, 17 October 2026

- Quirk-gen events through the ColliderML chain, in runs, multi-threaded, packaged as published ColliderML datasets
- Pilots at $\mu = 0$ and $\mu = 10$; simulation cost across the (m, Λ) grid
- An ACTS-based evaluator for quirk tracks; first GNN and MaskFormer results at $m = 200$ GeV, $\Lambda = 774$ eV

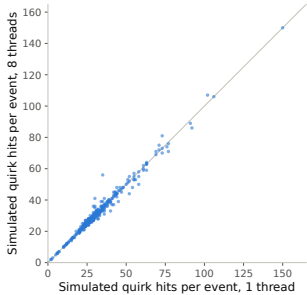
The chain

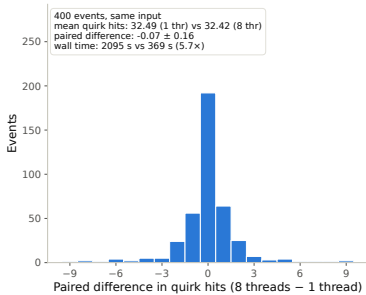


All stages run in one image, `ghcr.io/opendatadetector/sw:feat-python-pythia8-hepmc3`, with Docker, Podman, Apptainer or podman-hpc.
Stage scripts are colliderml-production's, called per run exactly as `run_stage.py` calls them:
`--output runs --output-subdir N --seed <dataset>_<version>_run<N>`

	colliderml-production	quirkml
layout	`<campaign>/<dataset>/<version>/runs/<N>`	same, campaign `quirks`, on CFS `simulation/`
seeds	hashed `<dataset>_<version>_run<N>`, every stage	same; quirk-gen gets the same integer
MadGraph	process built once, copied per run	quirk-gen's process built once, copied per run
parallelism	`threads` per run, `runs_per_node`	`sim_threads`, `--parallel K`, Slurm/HTCondor examples
packaging	`package_native_parquet.py` + verifier	same scripts, unchanged

QuirkDDG4 is now multi-threaded



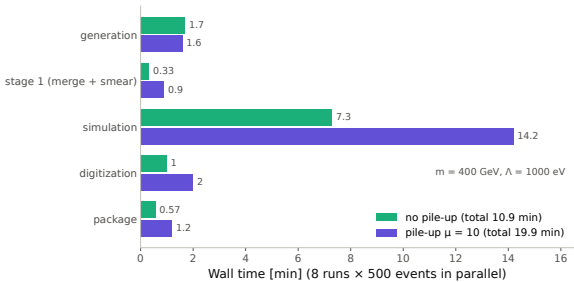


The ported Athena plugin kept each pair's string on the quirk particle definition, which Geant4 shares between worker threads, so it ran on one thread. Now the string state is per thread, the stacking action is installed on every worker, and strings are cleared at the start of each event.

$m = 200$ GeV, $\Lambda = 774$ eV, $\mu = 0$:

- 1 thread: 20/20 events identical to the old plugin
- 1 vs 8 threads, same 400 events: quirk hits/event 32.49 vs 32.42 (paired -0.07 ± 0.16); 2095 s vs 369 s
- plain ddsim without quirks is not reproducible event by event across thread counts either (11/40)

Pilots: 8 runs $\times$ 500 events at $\mu = 0$ and $\mu = 10$

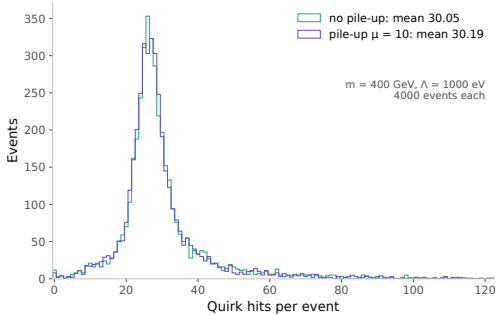


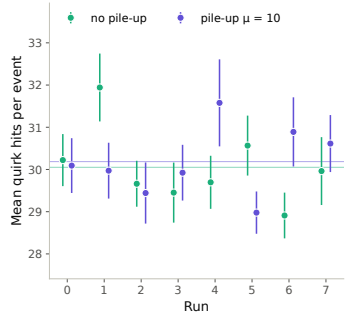
$m = 400$ GeV, $\Lambda = 1$ keV. Wall time per stage, 8 runs at once on one node. Simulation ran as one debug-queue job, one Slurm task per run (`colliderml/batch/slurm_runs.sh`).

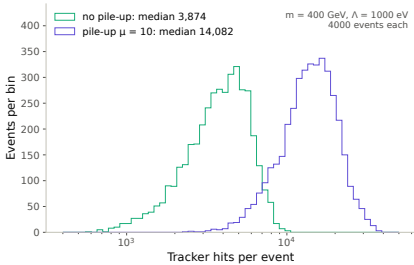
	$\mu = 0$	$\mu = 10$
events	4000	4000
unique ids and momenta	yes	yes
verifier	PASS	PASS
quirk hits / event*	30.05 ± 0.24	30.19 ± 0.26
tracker hits / event	3874	14082
sim. core-s / event	7	14
peak memory, 8 runs	n/a	65 GB
disk / event	1.7 MB	6.3 MB

* a hit counts if any of its truth particles is a quirk. Tracker hits: median.

Hits per event

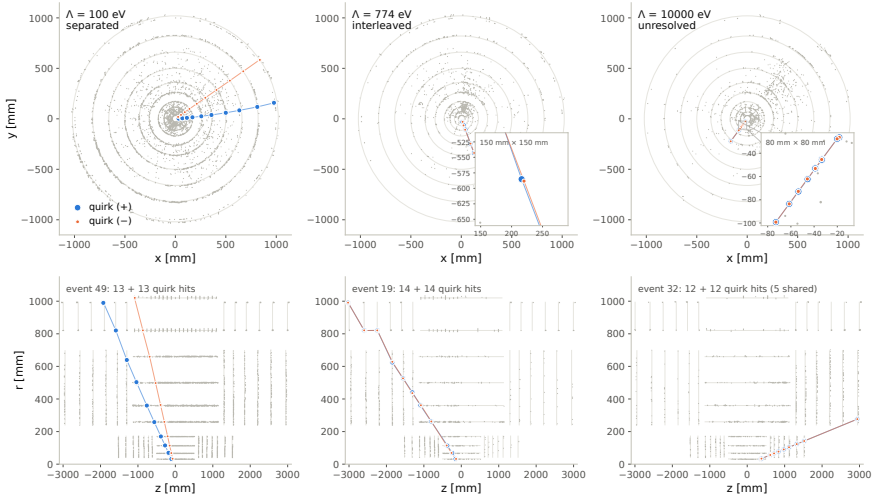






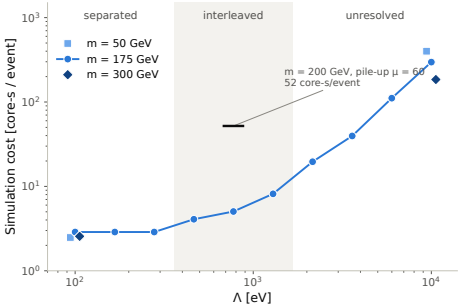
$m = 400$ GeV, $\Lambda = 1$ keV. Quirk hits do not change with pileup: $+0.13 \pm 0.35$ per event from $\mu = 0$ to $\mu = 10$. Per-run means are consistent within their errors.

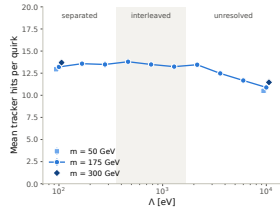
Three regimes of the string

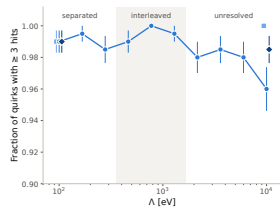


$m = 175$ GeV. Separated trajectories ($\Lambda \leq 278$ eV), interleaved (464-1292 eV), unresolved pair (≥ 2154 eV).

Simulation cost across the grid







- 15 points × 100 events, 8 threads per run, start-up removed.
- 3 core-s/event at $\Lambda \leq 278$ eV, 300 at 10 keV ($m = 175$ GeV), while hits per quirk barely change.
- At 10 keV mass matters: 400 ($m = 50$) and 185 ($m = 300$) core-s/event.

- Grid average at $\mu = 0$: about 51 core-s/event, so about 9000 events per node-hour. The 6 and 10 keV rows are about 80% of it.
- $\mu = 60$ at 774 eV: 52 core-s/event in simulation, 65 in all; 29 MB/event.

What about 1000 node-hours could produce

set	content	events	node-hours	disk
A	$\mu = 0$, every grid point (11×10), 20k each	2.2 M	∼245	4.2 TB
B	$\mu = 0$, $m = 200$ GeV, $\Lambda = 774$ eV, scaled to 1 M	1.0 M	∼20	1.9 TB
C	$\mu = 60$, every grid point, 2k each	220 k	∼55	6.4 TB
D	$\mu = 60$, 3 training points ($\Lambda = 278, 774, 2154$ eV), 100k each	300 k	∼50	8.7 TB
	total	3.7 M	∼370	21 TB

- Datasets go to `/global/cfs/cdirs/m4958/data/ColliderML/simulation/quirks/` (m4958 CFS: 303 of 512 TB used), as `<dataset>/<version>/{runs,parquet}`. The pilots, the 20k tracking set and the cost sets are moved there.
- CPU on m4958 (regular queue), GPU on m5386.
- Open: the mix of sets, and how much of the remaining ∼600 node-hours to spend now.

Tracking evaluation: ACTS, proto-track level

The ML-Tracking-Review evaluator (pyacts 48), copied into `quirkml/tracking/`. Metrics:

1. quirk track efficiency, per quirk (figure of merit): charged, primary, ≥ 3 hits, no p_T/η cut
2. quirk event efficiency: both quirks matched
3. technical efficiency ($p_T > 1$ GeV, $|\eta| < 3$, prompt, ≥ 7 hits), with and without quirks
4. fake and duplicate rates

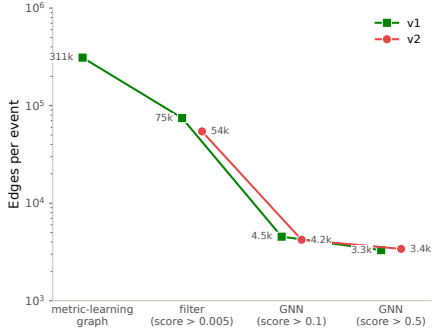
Matching: very loose (50% of hits, double matching) and tight (90%). The Kalman-fit level assumes helices: 0.6% quirk efficiency even for truth predictions, so it is not used.

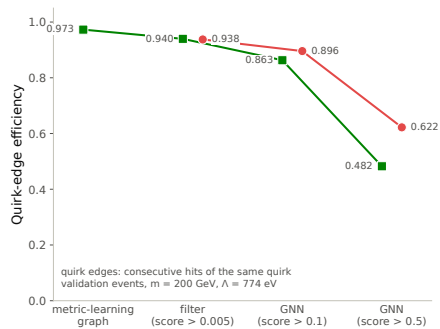
Validation on $m = 400$ GeV (4000 events; very loose / tight):

prediction	quirk	q. event	tech.	fake
truth	1 / 1	1 / 1	1 / 1	0
quirks split by time	1 / 0	1 / 0	1 / 0.90	0.002
60% quirk hits dropped	0	0	0.90	0.003
quirks kept with p = 0.5	0.50	0.24	0.95	0
nothing assigned	0	0	0	n/a

Truth predictions on the $m = 200$ GeV val and test ranges: 1 everywhere. 99.4% of quirks and 99.1-99.3% of events enter the denominators. An independent Python check agrees with ACTS in every event.

GNN chain: metric learning → filter → interaction GNN → track building

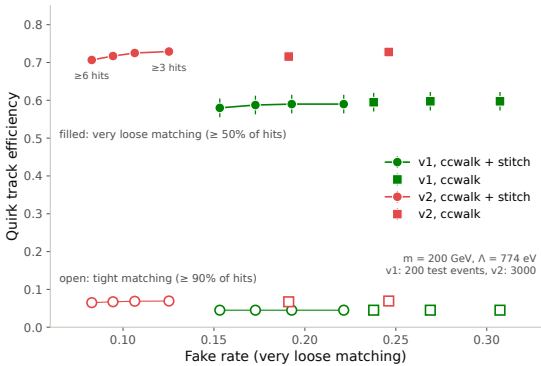




- Truth edges: each particle's hits ordered by true time (quirks turn back inwards, so radius order fails). Time is never an input.
- Inputs: position, volume, layer, and four cluster columns (sizes, channel count, summed activation).

- Embedding radius 0.07: 311k edges/event with 97.3% of quirk edges; after the filter, 94% remain at 54-75k edges/event.
- The GNN is the weak stage for quirks. The filtered graphs are saved for an edge-attention MaskFormer.

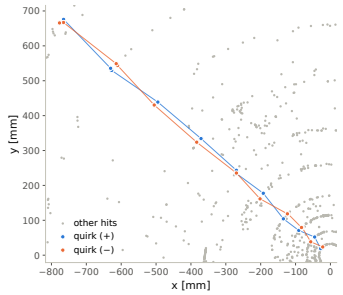
GNN results, $m = 200$ GeV, $\Lambda = 774$ eV, $\mu = 0$

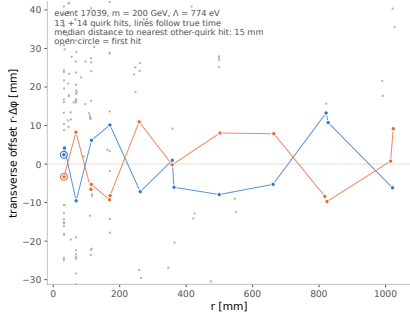


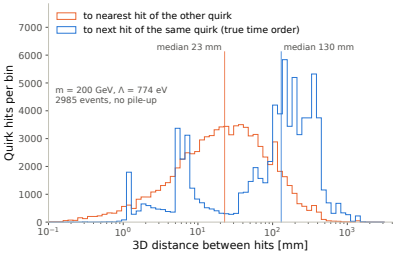
	v1	v2	v2
training events	4k	16k	16k
test events	200	3000	3000
track building	stitch, ≥ 5	ccwalk	stitch, ≥ 5
quirk eff. (50%)	0.588	0.728	0.717
quirk eff. (90%)	0.045	0.069	0.067
quirk event eff.	0.43	0.58	0.56
technical eff.	0.942	0.965	0.965
fake rate	0.173	0.246	0.095

v2 adds the cluster-feature differences on edges. Stitch: CC + walkthrough, then pieces joined when their end hits share an edge with score > 0.7, keeping tracks with ≥ 5 hits. The comparison against v2 without cluster features is still running.

Why quirk efficiency stops at about 0.7: the two quirks interleave



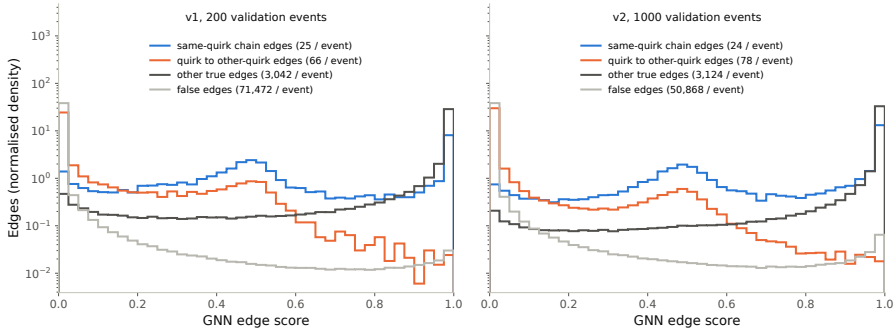




$m = 200$ GeV, $\Lambda = 774$ eV, 2985 test events. The other quirk's nearest hit is a median 23 mm away; consecutive hits of the same quirk are 130 mm apart. On the best v1 track per quirk, 81% of the wrong hits belong to the other quirk.

The pair is found (99% of pairs have half their hits on the union of v2 tracks). Separating the two quirks is the open problem at this point.

GNN edge scores for the two quirks



v1, 200 val events: same-quirk consecutive edges have median score 0.51 (51% above 0.5). Edges between the two quirks have median 0.007, but a bump at 0.4-0.5 of the same size as the true edges' (963 vs 943 edges). At each layer the next hit could belong to either quirk.

Hit direction might help here (not tested). In our parquet the cluster direction columns (`local_eta`, `local_phi`, `eta_angle`, …) are all zero, already in the per-run files written by ACTS.

MaskFormer (hepattn, from the LRT study)

- Ported to the quirk data: quirks are always targets (no p_T/η cut, sorted ahead of the 128-slot cut; at most 112 targets per event at $m = 200$ GeV).
- Logging: train and val loss, LR, grad norm, effective batch size, steps/s, GPU memory, to CSV and W&B.
- Overfit check: 8 events, decaying LR, 16/16 quirks and hit recall 1.0. Constant LR plateaued at 0.85-0.90.
- Throughput on the 20k set: 9.9 events/s at batch 1, 42 at batch 8 (10.6 GB).
- LR scan queued: batch 8, 30k steps, warmup-stable-decay, max LR 1.25×10^{-5}, 5×10^{-5}, 10^{-4}, 2×10^{-4}. Pending in the GPU queue since yesterday.
- Its predictions go straight into the evaluator: 4 memorised events gave 8/8 quirks.
- It attends over the whole event, so it allows us to test whether global context helps to separate the two quirks.

Things found along the way

- **Displaced vertices.** The Higgs-portal bug (ddsim drops status-22 long-lived particles and attaches their decay products to the primary vertex) does not affect quirks: every displaced generator particle in 100 $\mu = 0$ and 5 $\mu = 10$ events starts at its generator position after simulation, within 0.1 mm.
- **particle_id 0** is a real particle, often a quirk (21% of $m = 200$ GeV events). The first GNN cache treated it as noise; fixed and retrained.
- **Hits with two particles.** About 2.5% of hits carry more than one truth particle; the Release-1 layout keeps the first.
- **Quirks without hits.** About 0.7% of quirks have fewer than 3 hits.
- **ACTS evaluator.** By default ACTS drops particles below 1 GeV from its efficiency counts, and reads events by row order (rows within a file are not in event order). Both are handled in `evaluate.py`.
- **Cluster direction columns** are zero in the parquet (previous slide).

Next

1. GNN: finish the cluster-feature A/B (with / without), then separate the two quirks: hit direction once the writer fills it, and a pair-aware track builder.
2. MaskFormer: LR scan, then a long run; compare with the GNN at matched fake rate.
3. Edge-geo and flex-geo on the quirk-aware graph.
4. Generalisation: the trained models on other grid points without retraining, then training on more points.
5. Simulation: decide the mix of sets A-D, run on regular CPU (m4958), datasets on CFS.

Budget used: about 5 GPU-hours (m4958_g; m5386 from now on) and about 10 CPU node-hours.